STRhub Verified

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Technical Reproducibility Validation Report

GangSTR

Version v2.5

Illumina

Verification date	2026-11-08
Repository commit	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Result	**Does not run as documented: stops at install**
Reached	Available
Permalink	https://strhub.app/verified/gangstr
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Does not run as documented: stops at install**
REACHED	Available, 1/6 gates passed
WHY	The environment did not build from the declared install steps.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it does not run.

The environment did not build from the declared install steps. That does not mean the tool does not work. It means a third party following the README at this commit did not reach the end.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source **The run stopped here.**	FAIL
Starts	The installed program answers --help in the built environment	FAIL
Runs	Executes end-to-end without crashing *Not attempted, because the run had already stopped.*	—
Expected IO	Produces a non-empty file in the declared format *Not attempted, because the run had already stopped.*	—
Plausible Output	Output structure matches expected genotype-bearing format *Not attempted, because the run had already stopped.*	—
Execution log	gangstr.log-build.txt	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

4. Not Built From Source

STRhub tried to build the tool from its source at the pinned commit, following the build steps the repository declares, and the build failed. Nothing below the Installs gate ran.

What this means

– **If you are trying to run it:** A build from source at this commit fails in a clean environment; the cause and a suggested fix are below.

– **If you are reviewing a paper:** Nothing ran, so this says nothing about the software's output. It records that this attempt to build it stopped, and whose side the cause is on.

– **If you maintain it:** The cause could not be classified automatically. The full build output is linked below.

What failed

What happened	Suggested fix
An autotools build is missing its auxiliary file: config.sub	The configure script needs config.sub, which autoreconf -i (or automake --add-missing) copies in from the automake package. Running autoreconf -fi before ./configure in that step fixes it.

5. Errors Reported During the Run

The tool reported errors on 1 item(s) during the run. This does not assess whether the results produced are correct.

What happened	Times	Affected
An autotools build is missing its auxiliary file	1	config.sub

6. Terminal Error Output

Excerpt from the execution log when the run stopped:

```
[ 15%] Performing update step for 'htslib'
autoreconf: export WARNINGS=
autoreconf: Entering directory '.'
autoreconf: configure.ac: not using Gettext
autoreconf: running: aclocal --force
autoreconf: configure.ac: tracing
autoreconf: configure.ac: not using Libtool
autoreconf: running: /usr/bin/autoconf --force
configure.ac:32: error: required file 'config.sub' not found
configure.ac:32: try running autoreconf --install
autoreconf: error: /usr/bin/autoconf failed with exit status: 1
make[2]: *** [CMakeFiles/htslib.dir/build.make:101: thirdparty/htslib/src/htslib-stamp/htslib-update] Error
1
make[1]: *** [CMakeFiles/Makefile2:144: CMakeFiles/htslib.dir/all] Error 2
make: *** [Makefile:152: all] Error 2
```

7. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · GangSTR v2.5 · commit 6ea9b2b
GangSTR --bam /data/in/input.bam --ref /data/ref/hg38.fa --regions /data/in/regions.bed --out
/data/out/output
```

8. Reproducibility Metadata

Tool	GangSTR
Version	v2.5
Repository	https://github.com/gymreklab/gangstr
Commit (pinned)	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Container OS	ubuntu-22.04
Marker panel	Illumina
Verified on	2026-11-08 16:00:00 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42253585030

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

9. Output Content Evidence

No output files were generated because the run stopped before completion.

10. Verification Data

STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	—
License	—
Ref. genome	—
Loci tested	—

11. Verification Matrix

FAIL	Run on reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (0/5)	0/5 items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

12. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– A container environment, built from the tool declared install steps.

13. Scope and Limitations

Concerns reproducible execution only; no claim of genotype accuracy, concordance, forensic casework suitability, or regulatory validation.

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

- **Does not install**

  GangSTR v2.5 does not install in a clean ubuntu-22.04 environment at the pinned commit following the declared install steps (CMake build of bundled htslib fails).

- **Execution and output gates not reached**

  Because the tool did not install, execution on reference data was not attempted.

- **Verdict: Fails**

  The environment did not build from the declared install steps.

- **Reproducibility statement**

  The build instructions recorded in Section 3, executed at the pinned commit in the described environment, reproduce this failure.